

The State of Mid-Market Battery Storage

2026 Outlook — Synthetic Fixture

A 32-page sector outlook prepared for portfolio operators. All companies, projects, deal sizes, and individuals named in this document are **fictional** and exist only to give the html-anything PDF prompt a realistic shape to work against.

Prepared by Clockless Research · April 2026

Executive Summary

Mid-market behind-the-meter battery storage moved from a policy-dependent niche to a **capex-positive default** in 2025. On every project we modelled across the synthetic 47-site portfolio, front-of-meter pricing alone justified a four-hour battery; behind-the-meter demand-charge avoidance pushed simple payback inside seven years on 38 of 47 sites.

The structural change is **not** price decline of LFP cells, which has decelerated to 6.4% YoY. It is the collapse of soft costs — interconnection engineering, permitting, and commissioning — driven by software tooling that is now deployed at three of the four developers we surveyed.

We expect the mid-market segment to grow from 2.1 GWh of synthetic deployments in 2025 to 7.8 GWh by 2028, with the median project size declining from 1.9 MWh to 1.1 MWh as the developer ecosystem moves down-market.

Three findings drive the rest of this report:

1. Soft costs now dominate; cell price is a second-order lever. **2.** The bottleneck is utility interconnection queues, not capital. **3.** Operator software (DERMS) is the highest-margin layer and the most contested.

“The cheapest electron in 2026 is one that didn’t have to cross an interconnection.”

1. Market Structure

We define **mid-market** as projects between 250 kWh and 5 MWh of useable energy capacity, sited behind a commercial or industrial meter or in front of a distribution feeder serving 50–5,000 customers. This is the segment that historically had no native developer pool — too small for utility-scale shops, too large for residential installers.

1.1 Synthetic Portfolio Snapshot

The 47 fictional sites profiled in this report are distributed across nine U.S. states. Project sizes range from 480 kWh to 4.6 MWh. The table below summarises by state.

State	Sites	Capacity (MWh)	Median size (MWh)	Median IRR
California (synth.)	12	21.4	1.6	14.2%
Texas (synth.)	9	16.8	1.9	13.7%
New York (synth.)	6	8.1	1.3	11.4%
Massachusetts (synth.)	5	5.4	1.0	10.2%
Illinois (synth.)	4	3.9	0.9	9.6%
Other (synth.)	11	9.7	0.8	9.1%
Portfolio total	47	65.3	1.1	12.3%

1.2 Developer Concentration

Four fictional developers — **Halcyon Grid**, **Fieldline Energy**, **Mesa Pulse**, and **Bluemark Storage** — accounted for 71% of the synthetic 2025 deployment volume. Concentration is rising: in 2023 the same four held 49% share. We expect this to inflect downward in 2027 as the permitting tooling stack matures and lowers the entry bar.

“Halcyon’s real moat isn’t engineering, it’s that they ship 14-day permits in jurisdictions where the average is 90.” — synthetic operator interview, March 2026

2. Cost Structure

The shape of mid-market battery cost has flipped. As recently as 2022, cells were 56% of installed cost; by Q4 2025 they were 31%. The displaced share moved into engineering & permitting (now 22%) and commissioning (now 14%), both of which are labour-bound and therefore software-targetable.

Cost component	2022 share	2025 share	2028E share
LFP cells	56%	31%	26%
BMS & PCS	12%	13%	12%
EPC labor	10%	12%	11%
Engineering & permitting	9%	22%	16%
Commissioning	5%	14%	11%
Interconnection fees	5%	6%	12%
Developer margin	3%	2%	12%

2.1 The Software-Eatable Layer

Engineering & permitting, commissioning, and developer overhead together represent 38% of installed cost in 2025. This is the layer that is being compressed by operator software. Three of the four named developers above run an internal toolchain we collectively label DERMS-Plus: design automation, permit packet generation, commissioning checklist + telemetry, and ongoing dispatch optimisation.

Across the synthetic portfolio, sites built with DERMS-Plus tooling completed permitting in **17 days median** versus **83 days** for the legacy approach. The cost differential per project averaged **\$84,000**.

2.2 What is *not* being compressed

Interconnection fees are rising in three of the nine states profiled, driven by utility-side studies that price grid impacts more aggressively. We expect interconnection to *double* as a share of installed cost by 2028 even as everything else compresses. This is the next bottleneck.

3. Operator Software (DERMS)

Distributed Energy Resource Management Systems (DERMS) sit between the battery and the market signals it responds to. In the mid-market segment, DERMS choices fall into three buckets.

3.1 Three Architectures

Vendor-locked. The battery OEM ships the dispatch logic. Reliable but slow to adapt to new market mechanisms (e.g. ERCOT's synthetic *Ancillary Tier 3* product, hypothetically introduced 2026). Roughly 40% of the synthetic portfolio runs on this approach.

Independent SaaS. A third-party platform integrates with the OEM's API. Fastest to adapt, but introduces a contractual triangle. Roughly 35% of the synthetic portfolio.

Operator-built. The site operator (or its developer) writes the dispatch layer. Highest margin retention, highest engineering cost. Roughly 25% of the portfolio. Concentrated among the top two synthetic developers.

3.2 Where The Value Accrues

Across the synthetic portfolio, sites running operator-built DERMS captured **23% more revenue per MWh** versus vendor-locked, primarily because they could opportunistically arbitrage real-time prices and respond to demand-response calls without vendor mediation. The independent-SaaS bucket captured **17% more** than vendor-locked but trailed operator-built.

"Owning the dispatch logic is owning the cash flow. We give up on every other vertical integration argument before we give up on this one."

4. Risks & Counter-Theses

4.1 Interconnection Reform Stalls

FERC Order 2023 implementation across the nine modelled states is uneven; the worst-performing utility in the synthetic dataset has cleared 12% of its queue 18 months in. If reform stalls, the bottleneck identified in §2.2 hardens, and median IRRs across the portfolio fall **180–220 bps**.

4.2 Cell Tariff Volatility

Roughly 64% of the LFP cells in the synthetic portfolio originate from Asian manufacturers subject to current and pending tariff regimes. A 20-percentage-point tariff shock would move installed cost up roughly **6.2%** at the project level — meaningful but not portfolio-breaking.

4.3 Insurance Capacity

Battery insurance markets remain thin and concentrated. In 2025 three synthetic carriers wrote 84% of the policies on the portfolio. A single high-profile thermal incident could withdraw capacity industry-wide and delay 12–18 months of new builds. We rate this the highest-leverage low-probability tail in the report.

4.4 What we may have wrong

Our portfolio is synthetic and may overweight well-permitted jurisdictions. Real portfolios with heavier exposure to NIMBY-heavy counties will likely see higher soft costs and longer schedules. We have not modelled the impact of distribution-system upgrades that shift cost onto the developer rather than the utility.

5. Operator Recommendations

For mid-market portfolio operators evaluating where to allocate effort in the next 12–18 months, we recommend the following synthetic-portfolio-informed priorities:

5.1 Prioritise dispatch software ownership.

The 23% revenue uplift documented in §3.2 is the single largest controllable variable in the model. Operators who outsource dispatch are ceding the largest margin pool in the value chain. Build, buy with broad API access, or acquire — but do not lock in.

5.2 Move down-market deliberately.

The unit economics now support 480 kWh sites; the bottleneck is developer attention, not project economics. Operators with a permitting-automation toolchain are **structurally advantaged** at smaller sites.

5.3 Lock in interconnection upgrades early.

On the synthetic portfolio, sites whose interconnection studies completed before mid-2025 captured 41% lower fees than identical sites whose studies completed in Q4. Backlog growth means delay compounds; act ahead of the queue.

5.4 Diversify insurance.

Three-carrier concentration is unstable. Over the next 18 months, build relationships with at least two of the emerging European and synthetic-Lloyd's capacity providers identified in Annex C.

Glossary

BMS — Battery Management System. Electronics that monitor cells.

DERMS — Distributed Energy Resource Management System. Dispatch + monitoring software.

EPC — Engineering, Procurement, Construction. The on-site labour layer.

LFP — Lithium Iron Phosphate. The dominant mid-market cell chemistry.

PCS — Power Conversion System. The inverter between the battery and the grid.

FERC Order 2023 — U.S. federal rulemaking on interconnection queue reform.

Mid-market — projects between 250 kWh and 5 MWh useable capacity.

References

All references in this synthetic fixture are fictional. They mimic the shape of real industry citations (BloombergNEF, Wood Mackenzie, EIA reports, FERC dockets) so the html-anything PDF prompt has something to extract.

[1] Synthetic Energy Outlook 2026, fictional Halcyon Grid public deck, March 2026.

[2] Mock-FERC Docket ER26-118 (synthetic), interconnection queue review, January 2026.

[3] Fictional ERCOT Ancillary Tier 3 design memo, synthetic, February 2026.

[4] Imagined Mesa Pulse investor letter, Q4 2025 (synthetic).

[5] Fabricated Wood-Mackenzie-style storage cost teardown, January 2026.